Calibrated carry thresholds and smaller equations for ZFC consistency

Explicit nine-positive and ten-integer equations
with a new fifteen-positive degree–cost frontier

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Abstract

We combine independently useful advances from all eight incoming wave-7 archives, then improve their composition. A positivity-recovery bijection shifts only twelve of the thirty-one quadratic coordinates. A signed-borrow source uses sparse one-bit padding, and its exact binary mask population sets the central-binomial threshold. Calibrating that threshold separately for the Jones and Sun interfaces avoids an unnecessary degree penalty. A new quotient-cancelled radical majorant gives nine positive unknowns at 564 binary operations and exact degree 4 400 764, improving both coordinates of the best-degree reviewed incoming equation. A conditional three-root separation argument gives ten unrestricted integers at 554 operations and degree 14 699 152. A constant-cancelled norm minimum gives fifteen positive variables at 72 operations and degree 64, with several explicit tradeoffs. All equations bind the complete inherited 215492-byte native tuple, not a mixture of coefficient instances. The new proofs compose with identified inherited compiler, K215, Jones, and Sun theorems to give ZFC-provable unsolvability equivalences to the syntactic Con(ZFC) predicate. The archive contains literal arithmetic circuits, exact fixed payloads, source pins, mathematical proofs, and independently authored finite verifiers. These are not claims of global minimality, fewer unknowns, a consistency decision, or a new end-to-end proof-assistant certificate of these particular equations.

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1 Results and the precise comparison

Throughout $\mathbb{N} = \{0, 1, 2, \ldots\}$, $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$ and $\mathbb{Z}$ denotes all integers. The domain is part of an equation's specification.

Theorem 1.1 (Fixed consistency equations)**.** *Instantiate every fixed port and derived coefficient by the payloads and recipes in the accompanying release. For every retained circuit of its stated family, the mathematical arguments below and the identified inherited theorems establish in ZFC*

$$\begin{aligned}
\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{9}\ F_9(\mathbf{x}) = 0 &\iff \mathrm{Con(ZFC)},\\
\neg\exists \mathbf{x} \in \mathbb{Z}^{10}\ F_{10}(\mathbf{x}) = 0 &\iff \mathrm{Con(ZFC)},\\
\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{15}(\mathbf{x}) = 0 &\iff \mathrm{Con(ZFC)}.
\end{aligned}$$

The nine-nonnegative circuits give the same assertion over $\mathbb{N}^9$ by an explicit coordinate translation. No other domain conversion is implicit.

The consistency predicate means the absence of a finite syntactic ZFC refutation under the inherited K215 calibration. Proving this equivalence is not proving either side. In particular we infer no unconditional independence theorem merely from consistency.

Domain	Construction	Operations	Exact degree
$\mathbb{Z}_{>0}^{9}$	Incoming cheapest, previous iteration	548	15 502 748
$\mathbb{Z}_{>0}^{9}$	Incoming balanced, previous iteration	603	10 486 092
$\mathbb{Z}_{>0}^{9}$	Incoming lower degree, previous iteration	655	10 127 484
$\mathbb{Z}_{>0}^{9}$	Incoming Selective Quadratic	738	7 344 328
$\mathbb{Z}_{>0}^{9}$	New regular simple	490	9 461 300
$\mathbb{Z}_{>0}^{9}$	New regular simple	491	9 284 564
$\mathbb{Z}_{>0}^{9}$	New regular simple	494	8 923 436
$\mathbb{Z}_{>0}^{9}$	New packed simple	532	5 370 068
$\mathbb{Z}_{>0}^{9}$	New packed support	552	4 550 764
$\mathbb{Z}_{>0}^{9}$	New balanced	564	4 400 764
$\mathbb{Z}_{>0}^{9}$	New packed quotient	598	4 399 820
$\mathbb{Z}^{10}$	Incoming cheapest, previous iteration	582	50 822 936
$\mathbb{Z}^{10}$	Incoming balanced, previous iteration	637	34 378 296
$\mathbb{Z}^{10}$	Incoming Selective Quadratic	659	21 111 400
$\mathbb{Z}^{10}$	New lower cost	513	25 838 224
$\mathbb{Z}^{10}$	New lower cost	517	25 355 280
$\mathbb{Z}^{10}$	New lower cost	519	24 369 008
$\mathbb{Z}^{10}$	New balanced	554	14 699 152
$\mathbb{Z}^{10}$	New lower degree	558	14 696 016

Each displayed new low-variable point is nondominated within its delivered family. The incoming rows are the actual reviewed eight-archive frontier, not just a comparison against this report's own older points. Archive gate counts were independently recounted; inherited exact-degree claims retain their supplied mathematical proof boundary. The full review also records the other arrivals and their nonfrontier variants.

One paid operation means one live binary addition, subtraction or multiplication. Squaring is a multiplication; scalar multiplication is not free. Constants, input leaves and register reuse are free under this convention. All variable exponentiation and division are excluded from the literal evaluation graphs. These are arithmetic-circuit costs, not bit complexity, coefficient height, or expanded polynomial length.

The new balanced nine-positive equation reduces the best reviewed incoming nine-positive degree from 7 344 328 to 4 400 764, about 40.1%, and its operation count from 738 to 564, about 23.6%. The new balanced integer equation reduces 21 111 400 to 14 699 152, about 30.4%, while reducing 659 operations to 554, about 15.9%. These are meaningful finite-catalogue improvements, not claims of a worldwide or global optimum.

The eleven-positive 113/954 and twelve-positive 108/897 equations from the preceding iteration remain useful when trading a few more variables for low degree. This iteration does not improve those particular families. For fifteen positive variables, the short frontier below is often simpler than the low-variable equations if evaluation cost and total degree matter more than the number of unknowns.

1.1 What is new and what is inherited

1. **Selective positivity recovery.** Twelve guarded coordinates suffice. This is a proved zero-set bijection and reduces the encoding to 109 weighted records on 62 active monomials, without deleting unknowns.
2. **A calibrated source composition.** Sparse +1 target padding and the actual-popcount threshold come from the Selective Quadratic arrival. We compose them with a stronger selective source, interior padding, and distinct natural/integer threshold interfaces, proving all pre-decoding bounds and the signed-borrow cases.
3. **A lower-defect majorant.** Exact quotient cancellations and an additional finite radical estimate lower the polynomial error of the inherited T_6 majorant. The degree certificate applies to the actual primary norm polynomial, not a polynomial multiplied by a denominator power.
4. **Equal integer weights.** The incoming $(1, b, bK)$ separator can be shortened to $(1, b, b)$: after recovering the first square, the actual margin forces the remaining radicands to be distinct.
5. **Short positive equations.** The squared-modulus quotient and norm-minimum ideas combine; the extra cancellation $(B-1)^2 - 1 = B(B-2)$ then gives the new short-circuit frontier.

The complete native tuple with base 2^{429959} is inherited from the selective arrival. Its 215492-byte size is not a new record. What is new here is rebinding all ports and every derived leaf of these compositions and independently checking that binding. Search results are not optimality certificates, and a shorter arithmetic circuit is not necessarily a shorter binary description or expanded polynomial.

2 Source pin and logical boundary

The incoming-source pin is `4b4349064a750ba6347293b55874a7f92abdd0db`. All eight ZIP arrivals at this commit were read, their Git blob identities and ZIP CRCs verified, and their programs kept inert. Their shared construction baseline is `124c07c0b77a9e20e84676514d8b1ff78536ea3f`. The release records exact archive/member hashes but omits the full eight input ZIPs. Selected proof texts and necessary declarative data are retained. No historical archive number is invented; names and source paths identify the arrivals.

The proof chain is

$$\begin{aligned} \text{a zero of the fixed final polynomial} &\longleftrightarrow \text{a zero of the selective natural quadratic source} \\ &\longleftrightarrow \text{a zero of the positive quadratic parent} \\ &\longleftrightarrow \text{blank K215 halting} \\ &\longleftrightarrow \text{a finite syntactic ZFC refutation.} \end{aligned}$$

For the short positive family, its direct ordinary compiler proof replaces the selective-source step. We prove the new local transformations in ordinary mathematics formalizable in ZFC. The native compiler,

actual-machine calibration, Jones ratio theorem and Sun integer representation remain identified inherited dependencies. Exact finite checks substantiate the literal instances and algebra; they are not substitutes for the unbounded compiler theorems.

2.1 The complete quadratic parent

The 31 strictly positive coordinates, in certificate order, are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_a, \Delta.$$

The fixed ports are β, K_0, M_C, M_F. Define polynomial abbreviations

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha,$$
$$U = L + 1, \quad Y = a - E, \quad Q = c - \eta, \quad T = P - k, \quad R = jc - V.$$

The following are all 22 equations, in their canonical order:

$$\begin{aligned}
1:&\ U = q^2, & 2:&\ N = qU,\\
3:&\ X = wq, & 4:&\ Y = sN,\\
5:&\ E = XY, & 6:&\ Q = kY,\\
7:&\ T = EQ, & 8:&\ \tau^2 = PT + 1,\\
9:&\ D = X + ac + \gamma H, & 10:&\ \Delta = a^2 + H,\\
11:&\ C_2 = c^2, & 12:&\ D^2 = \Delta C_2 + 1,\\
13:&\ k = R + 1 + hE, & &\\
14:&\ R = (q - F)(N - q) - CL + (M_C + qM_F)J, & &\\
15:&\ (K_0 + w)C = F + z\beta J, & &\\
16:&\ F_2 = f^2, & 17:&\ V = of - c,\\
18:&\ g = y^2 - V^2, & 19:&\ I = iC_2,\\
20:&\ B_a = I^2, & 21:&\ B_a = \Delta(F_2 - 1),\\
22:&\ y^2 - B_a g = 1. & &
\end{aligned}$$

The residual is left minus right. The parent JSON is a literal shared sixty-operation evaluator, with SHA-256

`ac2a67782d344d1e74fd93bc3f09fae0097eaed1829fd8194ae235ae84763e48`.

Its old embedded instance label is historical: the port-parametric algebra is instantiated here by the complete new tuple. Its transport convention $z_{\mathrm{old}} = z + 1$ is valid on solution sets, not a bijection on arbitrary positive assignments. The authentic compiler gives $C > 0$, $F < q$, $X = wq = 2^R$ and $R > 3q$, hence $w > q > F$; therefore $(z_{\mathrm{old}} - 1)(q - 1) = (K_0 + w)C - F > 0$. Thus $z_{\mathrm{old}} \geq 2$, which justifies the inverse transport for the new tuple too.

2.2 The complete fixed coefficient instance

The native instance is `complemented-factorized-receiver-89x4831`. Its base is $B_{\mathrm{native}} = 2^{429959}$ and $\beta = B_{\mathrm{native}} - 1$. In port order (β, K_0, M_C, M_F) its bit lengths are

$$(429959, 434054, 429959, 429960).$$

The four minimal unsigned little-endian byte strings occupy respectively $53745, 54257, 53745, 53745$ bytes, totaling 215492. They are supplied as deterministic gzip payloads with both compressed and uncompressed SHA-256 hashes. The alias K_0 means the payload named K; M_F means `MF_source`, already shifted by $B_{\text{native}} - 1$. No second shift is permitted. The four ports are ordinary fixed integers, not existential parameters, unknown indices, or a search that waits for a consistency decision.

The inherited receiver adapter uses 457 selector modes, 17 receiver payload bits, 42 copied label bits, two standalone anchors and two dummy positions. Its key local repair complements the edge label code and uses a shared parity anchor to exclude nonzero payloads in empty cells. The full ordinary proof is `native/INHERITED_SEPARATED_PROOF.md`. Its historical opening says its own original delivery did not compose the new tuple into the main equations; this release explicitly does so for every active circuit.

Our independently authored finite reconstruction reads only the inert 430-action table and coefficient payloads. It checks 37960 basis identities, 1479 occupied decodings, 65076 single-bit changes, all 131072 empty payloads, and the exact 942 admissible tiles after unmatched/forbidden modes are removed. It also verifies mask populations, carry bounds, controller primality, multiplicative orders, lifting residues and every fixed byte. These establish the bounded instance facts, not the unbounded native-to-computation theorem.

The source's weighted coefficients satisfy

$$|c_{j,\alpha}| < 2^{864013} = M, \qquad \lambda = 2^{864026} = 8192M.$$

Every active derived leaf is materialized with an exact recipe over the four ports or as a fixed power of two. The literal graphs and fixed-binding files say how signed leaves are encoded. Thus a verifier can evaluate the specified polynomial without rebuilding the compiler or expanding its millions of degrees into a dense coefficient list.

3 Twelve guarded coordinates recover all positivity

Shift by one exactly the twelve parent coordinates

$$F, \alpha, z, h, i, j, s, w, \eta, \zeta, \gamma, g.$$

Represent the other nineteen coordinates by nonnegative integers directly. Let r_j denote the resulting quadratic residuals over $\mathbb{N}^{31}$.

Proposition 3.1. *This coordinate substitution is a bijection between the natural zero set of the transformed source and the strictly positive zero set of the parent. The transformed source has 109 coefficient records and 62 active monomials: 39 quadratic, 22 linear, and the constant. Its origin is not a zero.*

Proof. The twelve shifted coordinates are already positive. The transport row $(K_0 + w)C = F + z\beta J$ has a positive right side and positive multiplier, so $C > 0$. Consequently $q = F + \alpha + C \geq 3$ and $J > 0$. Rows 1–3 give $L = q^2 - 1 > 0$, $N = q(L + 1) > 0$ and $X = wq > 0$. Next $Y = sN > 0$, $E = XY > 0$, and $a = E + Y > 0$. Since $k = \eta + \zeta > 0$, the next rows give $Q = kY > 0$, $c = Q + \eta > 0$, $T = EQ > 0$ and $P = T + k > 0$; $\tau^2 = PT + 1$ makes the nonnegative τ positive. The defining rows give $D, \Delta, C_2 > 0$. Then $I = iC_2 > 0$ and $B_a = I^2 > 0$, while $B_a = \Delta(F_2 - 1)$ and $F_2 = f^2$ imply $f \geq 2$. Finally $y^2 = B_a g + 1 > 1$ makes $y > 0$, and

$$V^2 = (B_a - 1)g + 1 \geq 1$$

makes the nonnegative V positive. From $of = V + c > 0$ we get $o > 0$. This covers all nineteen initially unguarded coordinates. Conversely, subtract one from the twelve declared coordinates of any positive

zero. The two maps are literal inverses. At the natural origin row 2 is -1. Exact expansion of the listed residuals gives the stated support counts. □

This argument needs only $\beta, K_0 > 0$, not a compiler consequence, Pell-rank theorem, numerical specialization, or consistency assumption. It is a source-interface simplification, not a decrease in the number of quantified source variables. The bound and packing proofs that follow refer to this exact source, not the all-coordinate shift or a different transport convention.

4 Selective active support and freely oriented packing

The positivity-recovery reduction shifts only twelve coordinates, reducing 136 coefficient records to 109 and 71 active monomials to 62. The universal quadratic expansion still has 528 formal monomials. For padding position c, variable positions w_i, row signs $\epsilon_j \in \{\pm 1\}$, and targets H_j, define

$$m(\alpha) = (2 - |\alpha|)c + \sum_i \alpha_i w_i, \quad f(\alpha) = (2 - |\alpha|)! \prod_i \alpha_i!, \qquad D(T) = \sum_{j,\alpha} \epsilon_j f(\alpha) p_{j,\alpha} T^{H_j - m(\alpha)}.$$

The exact sparse isolation condition is

$$H_j - m(\alpha) + m(\gamma) = H_k \quad \Longleftrightarrow \quad j = k \text{ and } \alpha = \gamma,$$

for every active source record and every formal quadratic expansion label. It gives $[T^{H_j}]((T^c + \sum_i z_i T^{w_i})^2 D) = 2\epsilon_j q_j(\mathbf{z})$. It also forces all source exponents in D to be distinct, so overlapping bands incur no extra row-count factor in the coefficient bound.

A useful relaxation is to orient the row containing the highest source term rather than reserve a particular row as last. If that term's coefficient is a nonzero ordinary integer, its row sign can make it positive without changing support, isolation, or the common zero set. For $B > M + 1$, any positive integral leading coefficient dominates all lower coefficients bounded by M, proving $D(B) > 0$ unconditionally.

The smaller-K_{code} packed certificate has

$$(W, c, D_{\max}, K_{\mathrm{code}}) = (174, 81, 4072, 4420).$$

The shared-homogeneous certificate has

$$(W, c, D_{\max}, K_{\mathrm{code}}) = (170, 169, 4086, 4426), \qquad \min_j H_j = 340 = 2W.$$

The lowest final degree instead uses the unshifted $W = 170$ layout, with $K_{\mathrm{code}} = 4425$ and the odd threshold branch. Both displayed layouts have leading coefficient 1 and positive row signs. The latter requires only a uniform target shift by 1; it improves the previous shared-homogeneous $K_{\mathrm{code}} = 5262$ by approximately 15.9%. A smaller-width $W = 163$ regular certificate is retained for the operation frontier. All nine supplied variants verify $K_{\mathrm{code}} = D_{\max} + 2W \geq \max_j H_j + 2$, including the headroom needed by sparse one-bit padding.

Interior padding preserves the required witness bounds. At a nonnegative source zero, row 10 gives $\Delta = a^2 + 4a + 3 \geq 3$, while row 12 gives $D^2 = C_2\Delta + 1 \geq 1$, hence $D \geq 1$. Their distinct positions cannot both be zero, so the packed witness has a positive digit at a positive position and is at least $B \geq b$. The ordinary interval bounds are $B^c + g \leq 3bB^W$ or $5bB^W$, according to the bridge's witness interval.

A newly authored standard-library verifier reconstructs the complete selective source from pinned inert gate data, checks every coefficient and row sign, and exhaustively replays 57 552 positional checks per certificate. Search code is not a proof dependency; global optimality and new kernel formalization of the inherited compiler endpoint are not claimed.

4.1 Literal positional data

The ordered positions below use the 31-coordinate order of the quadratic parent. The complete nine JSON certificates also give every reversed coefficient and its exponent, so these lists and the source equations uniquely specify the positional codes.

Operation-oriented regular layout $W = 163$, $p = 0$, $d = 7194$, $K_c = 7520$; negative row signs: 22.

$$(w_1, \dots, w_{11}) = (8, 30, 44, 87, 158, 90, 151, 163, 135, 147, 7)$$
$$(w_{12}, \dots, w_{22}) = (150, 103, 80, 136, 67, 42, 156, 130, 17, 3, 79)$$
$$(w_{23}, \dots, w_{31}) = (162, 128, 18, 115, 56, 141, 46, 91, 140).$$

$$(H_1, \dots, H_{11}) = (327, 654, 981, 1308, 1635, 1962, 2289, 2616, 2943, 3270, 3597)$$
$$(H_{12}, \dots, H_{22}) = (3924, 4251, 4578, 4905, 5232, 5559, 5886, 6213, 6540, 6867, 7194).$$

Balanced shared-homogeneous layout $W = 170$, $p = 169$, $d = 4086$, $K_c = 4426$; all row signs are positive.

$$(w_1, \dots, w_{11}) = (8, 11, 6, 42, 0, 112, 4, 72, 116, 3, 24)$$
$$(w_{12}, \dots, w_{22}) = (1, 150, 78, 9, 80, 52, 111, 170, 163, 139, 159)$$
$$(w_{23}, \dots, w_{31}) = (37, 40, 87, 83, 118, 114, 146, 155, 45).$$

$$(H_1, \dots, H_{11}) = (383, 1219, 3528, 2539, 3833, 1063, 340, 724, 4292, 3807, 2196)$$
$$(H_{12}, \dots, H_{22}) = (2290, 1780, 1441, 2743, 3951, 2086, 3055, 3300, 721, 3084, 1864).$$

Degree-oriented odd-threshold layout $W = 170$, $p = 169$, $d = 4085$, $K_c = 4425$; all row signs are positive.

$$(w_1, \dots, w_{11}) = (8, 11, 6, 42, 0, 112, 4, 72, 116, 3, 24)$$
$$(w_{12}, \dots, w_{22}) = (1, 150, 78, 9, 80, 52, 111, 170, 163, 139, 159)$$
$$(w_{23}, \dots, w_{31}) = (37, 40, 87, 83, 118, 114, 146, 155, 45).$$

$$(H_1, \dots, H_{11}) = (382, 1218, 3527, 2538, 3832, 1062, 339, 723, 4291, 3806, 2195)$$
$$(H_{12}, \dots, H_{22}) = (2289, 1779, 1440, 2742, 3950, 2085, 3054, 3299, 720, 3083, 1863).$$

5 Sparse padding and the exact carry threshold

Let p be the padding position, w_i the 31 distinct variable positions, $W = \max(p, w_i)$, and H_j the 22 distinct targets. Let $D_c(t)$ be the weighted, row-oriented source polynomial of the selected certificate, $d = \deg D_c$, $K_c = d + 2W$, and $E = K_c + W + 2$. Every reversed exponent is nonnegative. The certificate proves

$$[t^{H_j}] \left(t^p + \sum_i x_i t^{w_i} \right)^2 D_c(t) = 2\epsilon_j r_j(\mathbf{x}), \qquad K_c \geq \max_j H_j + 2.$$

The maximum target, source degree, and padding position are distinct data. The 109 reversed positions are distinct, so a bound on each weighted source coefficient also bounds each coefficient of D_c, without a row-count factor.

Put $\kappa = 864026$, $\lambda = 2^\kappa$, and $B = \lambda b^2$ for integer $b \geq 1$. The positive integral leading coefficient of D_c implies

$$0 < D_c(B) < \frac{MB^{d+1}}{B-1}, \qquad M = 2^{864013},$$

because all lower terms together have magnitude below $MB^d/(B-1) < B^d$. This positivity is unconditional; it does not assume a source zero or a binary base.

5.1 The sparse source and its size bounds

For $g \geq 0$, define only polynomial abbreviations

$$\begin{aligned} A_c &= \sum_j B^{H_j}, & \ell &= \sum_i B^{w_i}, & N_0 &= B^{W+1}, \\ Z_c &= B^{K_c+1}, & M_c &= N_0 - 1 - (b-1)\ell, & & \\ J_c &= 2(B^p+g)^2 D_c(B) + A_c, & & & & \\ S_c &= g + N_0 J_c, & T_c &= M_c + N_0(B-2)A_c. & & \end{aligned}$$

The ordinary radix-B digits of M_c are $B-b$ at a variable position, and $B-1$ elsewhere through W. Thus $0 \leq M_c < N_0$, $T_c \geq 0$, and $J_c, S_c > 0$, before b is known to be a power of two.

On the branch $g < 4bB^W$ one has $g < N_0$ and $B^p + g < 5bB^W$, so

$$2(B^p+g)^2 D_c(B) < \frac{50Mb^2B^{K_c+1}}{B-1} \leq \frac{50}{8192}\frac{BZ_c}{B-1} < \frac{Z_c}{128}.$$

Also $A_c < B^{\max H_j+1}/(B-1) < Z_c/128$, whence $0 < J_c < Z_c/64$. Finally $(B-2)A_c < B^{\max H_j+1} \leq Z_c$. Integer low/high block bounds give

$$0 \leq S_c, T_c < B^E, \qquad S_c + T_c < 2B^E. \tag{C1}$$

Using $1+g$ in place of B^p+g would invalidate the proof for interior padding. The displayed estimate explicitly retains p.

5.2 Signed borrows rather than fictitious positive digits

Now assume $b = 2^k$ and hence $B = 2^v$, $v = \kappa + 2k$. The low mask condition $g \,\&\, M_c = 0$ is exactly

$$g = \sum_i x_i B^{w_i}, \qquad 0 \leq x_i < b.$$

The padding position is distinct, so its digit remains the fixed coefficient one in the encoded linear form. Write h_t for the coefficient of $(t^p + \sum_i x_i t^{w_i})^2 D_c(t)$. Then

$$|h_t| < M(1 + \sum_i x_i)^2 \leq M(31b)^2 < 1024Mb^2 = B/8.$$

The nominal digit of J_c is $2h_t + \delta_t$, where $\delta_t = 1$ at a target and is zero elsewhere. Starting with incoming carry $c_{-1} = 0$, normalize by

$$c_t = \left\lfloor \frac{2h_t + \delta_t + c_{t-1}}{B} \right\rfloor.$$

The strict coefficient bound proves inductively $c_t \in \{-1, 0\}$. At a target, $h_t = 0$ gives normalized digit 0 or 1. If $h_t > 0$, the digit is at least 2 and below $B/2$; if $h_t < 0$, it is above $B/2$. Because every h_t

at a target is $2\epsilon_j r_j$, a nonzero positive h_t is at least 2, which supplies the first lower bound even after a borrow. Precisely the first case is disjoint from the target mask $B-2$. As $0 < J_c < Z_c$, these are ordinary digits of a finite nonnegative integer; no bit convention for a negative integer is being used. Hence

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ encodes a common natural source zero.} \tag{C2}$$

5.3 A two-block valuation identity

For $S, T \geq 0$ with $S + T < L = 2^u$ and fixed integer $c > 0$, put

$$R_c = cL^2 + (S+T)L + (L-1-S).$$

These three binary blocks do not overlap. Complementing the low block gives

$$\mathrm{pc}(R_c) = \mathrm{pc}(c) + u + \mathrm{pc}(T) - \delta, \qquad \delta = \mathrm{pc}(S) + \mathrm{pc}(T) - \mathrm{pc}(S+T) \geq 0.$$

Here $\delta = 0$ exactly when $S \,\&\, T = 0$; it counts the propagated binary carries. Legendre's factorial formula gives $v_2\binom{2R_c}{R_c} = \mathrm{pc}(R_c)$, so

$$2^{u+\mathrm{pc}(T)+\mathrm{pc}(c)} \mid \binom{2R_c}{R_c} \quad \Longleftrightarrow \quad S \,\&\, T = 0. \tag{C3}$$

It is essential to use $S + T < L$, not just two separate upper bounds. For our mask the 31 variable positions remove k one-bits each, while the 22 high targets contribute $v - 1$ each. Thus

$$\mathrm{pc}(T_c) = (W+23)v - 31k - 22. \tag{C4}$$

5.4 The nine-variable square threshold

Jones's interface needs a threshold of the form N^2. Set

$$L = 2bB^E, \quad R = L^2 + (S_c + T_c)L + L - 1 - S_c, \quad G_{\mathrm{cap}} = 32bB^W.$$

By (C1), $S_c + T_c < L$. Here $u = 1 + k + Ev$, and the top word is $c = 1$. Equations (C3)–(C4) give threshold exponent

$$(K_c + 2W + 25)v - 30k - 20. \tag{C5}$$

For even K_c, take

$$q = (K_c + 2W + 10)/2, \qquad N = 2^{15\kappa/2-10}B^q.$$

For odd K_c, take instead

$$q = (K_c + 2W + 9)/2, \qquad N = 2^{8\kappa-10}bB^q.$$

In both cases $2\log_2 N$ equals (C5) when $b = 2^k$. Both expressions are polynomials with fixed integer exponents and coefficients. The odd branch is required by the degree-minimal $W = 170$, $K_c = 4425$ certificate. The square-root threshold N is *not* the carry block length L.

For all initial assignments $b, g \geq 1$, without a binary-base premise,

$$R = L^2 + (L-1)S_c + LT_c + L - 1 \geq L^2 + L - 1 \geq 8, \qquad N \geq \max(b, 8), \quad R \geq b, \quad G_{\mathrm{cap}} > 0.$$

The imported ratio theorem first supplies $8g < G_{\mathrm{cap}}$, which is exactly the branch used to prove (C1). Conversely a source zero can be encoded for every sufficiently large power of two b; it is nonzero, so $g \geq 1$, and distinct positions give $g < bB^W$, hence $32g < G_{\mathrm{cap}}$. The endpoint $b = 1$ produces no extra zero: its mask forces every decoded source digit to zero, contrary to residual 2 at the origin.

5.5 The integer threshold must remain different

The Sun bridge allows an arbitrary polynomial threshold Y. Use

$$L = B^E, \quad X = L^2 + (S_c + T_c)L + L - 1 - S_c, \quad Y = 2^{\kappa(K_c+2W+25)-21} b^{2(K_c+2W+25)-31}, \quad \mathcal{C} = bB^W.$$

On its soundness branch $f \neq 0$, $b = 1 + 3f$ is a square, so $b \geq 4$, and $0 \leq g < 2\mathcal{C}$. The certificate's target gap is at least two, giving $(B-2)A_c < B^{K_c-1} = Z_c/B^2$. Together with $J_c < Z_c/64$ and $(g + M_c)/N_0 < 2 < Z_c/128$, this gives the stronger bound $S_c + T_c < L$. The exact threshold exponent is now

$$(K_c + 2W + 25)v - 31k - 21,$$

which is exactly $\log_2 Y$. Moreover even κ and $b \equiv 1 \pmod 3$ give $B, L \equiv 1 \pmod 3$ and $T_c \equiv -22 \equiv 2 \pmod 3$; therefore $X \equiv 1 + T_c \equiv 0 \pmod 3$ before binary-base recovery. The bounds $X > 3b$ and $Y > \max(b, 256)$ are immediate.

Completeness needs $b \leq g < \mathcal{C}$. At a source zero the distinct coordinates D, Δ are positive and at most one of their positions is zero, so $g \geq B \geq b$. A sufficiently large power-of-two square supplies the remaining bound. Copying the natural choice $L = 2bB^E$ into this branch would needlessly change the degrees and can destroy $3 \mid X$; the two interfaces are deliberately kept separate.

6 The nine variable ratio interface

We first use exactly the mixed-domain coordinates

$$b, g, h, s, w, i \in \mathbb{Z}_{>0}, \qquad \phi, j, n \in \mathbb{N}.$$

Every other symbol is a polynomial abbreviation. The source registers $R, N, G_c = G_{\mathrm{cap}}$ are those of Section 5, with the selected even- or odd-K_c threshold formula. They satisfy the imported ratio theorem's unconditional size requirements before a zero is assumed. The following quotient removal and conditional norm lemma are retained from the preceding iteration; the new source and sharper majorant are composed with their full two-way proofs, not merely with sample tests.

6.1 Literal ratio registers and exact quotient removal

Define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R + 1 + h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R + 1, & W_1 &= bw, & C &= B_1 + W_1 + \phi,\\
J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & &\\
D &= (A^2-1)C^2 + 1, & k &= iJC, & L &= kD,\\
E &= LC, & F &= (A^2-1)E^2 + 1, & &\\
G &= A + F(CD + 1 - A), & H &= C + F(B_1 + jC), & &\\
I &= (G^2-1)H^2 + 1, & a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI,\\
m &= [K^2 - 4(C - KY)^2]G_c - 8gK^2. & & & &
\end{aligned}$$

The inherited ratio theorem instead has $H_{\mathrm{old}} = B_1 + j_{\mathrm{old}} C$, $j_{\mathrm{old}} \geq 1$, and divisibility $JF \mid QF + J(H_{\mathrm{old}} - C)$. Since $F \equiv 1 \pmod J$ and $F \equiv 1 \pmod C$, $\gcd(F, J) = 1$, so that divisibility is equivalent to

$$J \mid Q \quad \text{and} \quad F \mid H_{\mathrm{old}} - C.$$

If the second holds, $z = (H_{\text{old}} - C)/F$ is positive and congruent to B_1 modulo C. As $0 < B_1 < C$, uniquely $z = B_1 + jC$ for $j \geq 0$. This is exactly the new H. Conversely every such new j gives the old positive integer

$$j_{\text{old}} = 1 + \frac{F-1}{C} B_1 + Fj.$$

The quotient is an integer because $C \mid F - 1$. This is a bijection of the old divisible witnesses with the new unrestricted nonnegative witnesses, preserving H itself. No cofinal-witness assertion or extra hypothesis is introduced. The resulting ratio conditions are precisely

$$a_0 \text{ square}, \quad a_1 \text{ square}, \quad J \mid Q, \quad m > 0.$$

The numerical Jones theorem gives from these conditions that b is a power of two, $8g < G_c$, and $N^2 \mid \binom{2R}{R}$. Conversely the stronger source bound $32g < G_c$ produces the requisite ratio witnesses.

6.2 A general conditional-separation norm theorem

Theorem 6.1. *Let $u, v, m, a_0, a_1, V_0, V_1$ be integers with $u > 0$, $0 < v/u < V_0 + V_1$, $a_i > 0$, $V_i > 0$, $a_i < V_i^2$, and $m > 0 \Rightarrow a_0 \neq a_1$. Put*

$$\Omega = um, \qquad \Theta = un + v - \Omega(V_0 + V_1).$$

Then there exists $n \in \mathbb{N}$ satisfying

$$(\Theta^2 - \Omega^2(a_0 + a_1))^2 - 4\Omega^4 a_0 a_1 = 0$$

if and only if $m > 0$, a_0, a_1 are integer squares, and $u \mid v$.

Proof. If $m = 0$, $\Theta = un + v > 0$, excluding a zero. Otherwise factor the norm over the reals to obtain signed roots $r_i^2 = a_i$ with $\Theta/\Omega = r_0 + r_1$. Rearrangement gives

$$n + v/u = mZ, \qquad Z = V_0 + V_1 + r_0 + r_1 > 0.$$

The left side is positive, so $m > 0$ *before* integrality of either root is known. Conditional separation now makes $x = r_0 + r_1$ nonzero; otherwise $a_0 = a_1$. Since x is rational, $2xr_0 = x^2 + a_0 - a_1$ shows $r_0, r_1 \in \mathbb{Q}$. A rational number whose square is an integer is an integer, as follows by writing it as a reduced fraction. Hence $Z \in \mathbb{Z}$ and $u \mid v$. Conversely choose the positive integer roots and put $n = mZ - v/u$. Then $n \in \mathbb{Z}$ and $n \geq Z - v/u > V_0 + V_1 - v/u > 0$. Substitution makes a linear factor zero. The quotient bound is needed only for completeness, not for the positivity-first soundness argument. □

Positive majorants are essential, not just $V_i^2 > a_i$. Without separation, $a_0 = a_1 = 2$, $V_0 = V_1 = 2$, $u = v = 2$, $m = 1$, $n = 3$ is a false square witness. Without positive majorants, $a_0 = 1, a_1 = 4, V_0 = -2, V_1 = 4$, $u = v = 1, m = -1, n = 0$ is a false positive-margin witness, even though the squared-majorant and quotient inequalities hold. These are retained as negative controls.

6.3 The lower cost simple majorant companion

Unconditionally $A \geq 2$ and $P, K, C, J, D, E, F, G, H > 1$. Define

$$V_0 = PK, \qquad T = k(8D^2 - 4D - 1), \qquad V_1 = TGH.$$

The identities $E = CL$, $L = kD$ give $F = (D-1)k^2D^2 + 1$. Direct expansion then yields

$$V_0^2 - a_0 = K^2 - 1 > 0, \qquad T^2 - 64DF = (8D+1)k^2 - 64D = 8D(k^2 - 8) + k^2 > 0,$$

because $k = iJC \geq 3$. Therefore the strictly positive majorant V_1 has

$$\Delta_1 = V_1^2 - 64a_1 = [(8D+1)k^2 - 64D](GH)^2 + 64DF(H^2-1) > 0.$$

The lower degree of this polynomial defect, rather than simply a smaller numerical bound, is the degree improvement.

If $m > 0$, its defining margin implies $4(C - KY)^2 < K^2$, so $C > K(Y - 1/2) > K$ because $Y \geq 64$. Also

$$A^2 - P = M_0^2(U^2+1) > 0, \quad D > C^2, \quad F \geq A^2, \quad G \geq A, \quad I \geq G^2 \geq A^2.$$

For $G \geq A$, use $G = A + F(CD + 1 - A)$ and $D \geq A^2$, $C \geq 1$. It follows that

$$a_1 = DFI > C^2A^4 > C^2P^2 > K^2P^2 > a_0.$$

This separation is established without assuming squarehood or divisibility. The same positivity-first proof works without the auxiliary quotient offsets. Indeed $Q < 3C^2$, $D > C^2$ and $T > 3D$ imply $V_1 > Q$. The lower-cost simple-majorant fixed polynomial is

$$\begin{aligned} \Omega &= Jm, \\ \Theta &= Jn + Q - \Omega(V_0 + V_1), \\ F_9 &= [\Theta^2 - \Omega^2(a_0 + 64a_1)]^2 - 256\Omega^4 a_0 a_1 = 0. \end{aligned}$$

To check the offset-free form explicitly, $m = 0$ gives $\Theta = Jn + Q > 0$. Otherwise a zero factor gives $n + Q/J = mZ$ with $Z = V_0 + V_1 + r_0 + r_1 > 0$. Thus $m > 0$, and the separation just proved makes $r_0 + r_1$ nonzero. The rational reconstruction from Theorem 6.1 gives integer roots and $J \mid Q$. Conversely choose the positive integer roots and set $n = mZ - Q/J$. As $Z > V_1 > Q > Q/J$, this is a positive integer and a factor vanishes.

Squarehood of $64a_1$ is equivalent to squarehood of a_1: an integer square root divided by 8 is a rational square root of an integer, hence an integer. The finalizer is therefore equivalent to exactly the four ratio conditions. Composing both directions proves the desired source solvability equivalence.

For the positive-domain polynomial, replace ϕ, j, n in this mixed-domain formula by the corresponding positive coordinates minus 1. For the nonnegative-domain polynomial, replace b, g, h, s, w, i by the corresponding nonnegative coordinates plus 1. These are bijections; the supplied circuit ledgers charge their operations. They must not be omitted when stating the domain.

7 A sharper polynomial radical bound

The following local improvement is independent of the choice of sparse source. Keep the CRT-reparameterized Jones registers, and write

$$k = iJC, \quad J = 4A - 5, \quad D = (A^2-1)C^2 + 1, \quad \mathcal{L} = CD + 1 - A.$$

Then $F = 1 + k^2D^2(D-1)$, $G = A + F\mathcal{L}$, and $I = (G^2-1)H^2 + 1$. All allowed assignments have $A \geq 33280$, $C \geq 1$, and $i \geq 1$; the argument below only requires $A \geq 729$. These letters are polynomial registers, not additional unknowns.

Put

$$B_8 = 8D^2 - 4D - 1, \qquad \mathcal{N} = 2G + \mathcal{L},$$
$$T_6 = k\{128B_8\mathcal{N} - 8k^2(2B_8 + 2D - 1)\mathcal{L} + C(14k^2 - 192)\},$$

$$T = T_6 + i^3\{9C^2(64A - 239) - 14C(64A^2 - 304A + 603) + 1024\} + 288i + 2048,$$
$$V_1 = TH, \qquad q = 2048^2.$$

The supplied support-majorants construction already used T_6 with a coarser positive correction. The new step is the correction displayed here: exact polynomial division in A, together with one further finite radical estimate, removes its high-degree error terms.

Uniform majorant. For every allowed assignment,

$$T > 2048\sqrt{DF}\,G > 0, \qquad V_1 > \sqrt{qDFI}. \tag{1}$$

To prove this, define

$$t_7 = 2048D^7 - 1024D^6 - 256D^5 - 128D^4 - 80D^3 - 56D^2 - 42D - 33.$$

Exact multiplication gives

$$t_7^2 - 2048^2 D^{13}(D-1) = 109824D^6 + 36608D^5 + 18304D^4 + 9984D^3 + 5460D^2 + 2772D + 1089 > 0.$$

Also $t_7 > 0$ for $D \geq 1$, since $2048 > 1719$, the sum of the absolute values of its other coefficients. Thus $t_7/(2D^5) > 1024D^2\sqrt{1 - 1/D}$. No convergence argument is used.

A second exact identity is

$$T_6 = \frac{kt_7\mathcal{N}}{2D^5} + \mathcal{E}_7,$$

where

$$\mathcal{E}_7 = \frac{14k^3(A-1) - 9k^3C}{D} + \frac{9k^3(A-1) - 33k^3C}{D^2} + \frac{33k^3(A-1)}{D^3} + k\left\{\frac{120C + 192 - 64A}{D} + \frac{84C + 120 - 40A}{D^2} + \frac{63C + 84 - 28A}{D^3} + \frac{63 - 21A}{D^4}\right\} + \frac{99kC}{2D^4} + \frac{33k(3-A)}{2D^5}.$$

The decisive divisions are

$$J^3 = (A+1)(64A^2 - 304A + 604) - 729,$$
$$J^3 = (A^2 - 1)(64A - 240) + (364A - 365).$$

Writing $s = A^2 - 1$ and using $D = sC^2 + 1$, they imply

$$\frac{14k^3(A-1)}{D} - 14i^3C(64A^2 - 304A + 604) = -14i^3C\left\{\frac{729}{A+1} + \frac{J^3}{(A+1)D}\right\},$$
$$-\frac{9k^3C}{D} + 9i^3C^2(64A - 240) = -\frac{9i^3C^2(364A - 365)}{s} + \frac{9i^3J^3C^2}{sD}.$$

For $A \geq 729$ and $C \geq 1$, the elementary bounds

$$\frac{729}{A+1} < 1, \quad 0 < \frac{364A - 365}{A^2 - 1} < 1, \quad \frac{J^3}{(A+1)D} < \frac{64}{C^2}, \quad \frac{k^3C}{D^2} < i^3, \quad \frac{kA}{D} < 4i$$

show, on dropping positive summands, that

$$\mathcal{E}_7 - 14i^3C(64A^2 - 304A + 604) + 9i^3C^2(64A - 240) > -14i^3C - 9i^3C^2 - 929i^3 - 277i.$$

Here $929 = 14 \cdot 64 + 33$. The bound $277i$ follows from $D \geq 9$, $64 + 40/9 + 28/81 + 21/729 < 69$, and $66/D^4 < 1$. Consequently $T > kt_7\mathcal{N}/(2D^5) + 2048$.

Finally let $X = k^2D^3(D-1)$. Then $DF = X + D$ and $G = (X/D)\mathcal{L} + CD + 1$. Strict concavity of the square root gives

$$2048\sqrt{DF}\,G < 1024kD^2\sqrt{1 - 1/D}\,(2G + \mathcal{L}) + 2048v, \qquad v = \frac{CD+1}{2kD\sqrt{1-1/D}} < 1.$$

The last inequality follows from $k \geq 7C$, $D \geq 9$, and $\sqrt{D(D-1)} > D - 1$. This proves the first inequality in (1); the second follows from $I < (GH)^2$. Every denominator above occurs only in the proof, not in the polynomial.

The offset-free finalizer. Let $a_0 = (P^2 - 1)K^2 + 1$, $a_1 = DFI$, $V_0 = PK$, and retain

$$m = [K^2 - 4(C - KY)^2]G_{\mathrm{cap}} - 8gK^2, \qquad \Omega = Jm.$$

For the nonnegative witness n, put

$$E_0 = Jn + Q, \qquad \Theta = E_0 - \Omega(V_0 + V_1),$$

so the final equation is

$$\boxed{\left[\Theta^2 - \Omega^2(a_0 + qa_1)\right]^2 - 4q\Omega^4 a_0 a_1 = 0.} \tag{2}$$

Its zero predicate is exactly the four ratio conditions a_0, a_1 square, $J \mid Q$, and $m > 0$. Indeed, $m = 0$ would leave $(Jn + Q)^4 > 0$. Otherwise real factorization of the norm gives

$$n + Q/J = m(V_0 + V_1 + r_0 + r_1), \qquad r_0^2 = a_0, \quad r_1^2 = qa_1.$$

Both majorants are unconditional, so the parenthesized expression is positive and $m > 0$ follows before any integrality assertion. The margin then implies $C > K$, and the standard ratio inequalities give $a_1 > a_0$. The rational signed root sum therefore cannot vanish. Squaring isolates each root, proving it rational and hence integral; also $r_1/2048$ is an integral square root of a_1. The displayed equation now gives $J \mid Q$. Conversely choose the positive integral roots and set $n = m(V_0 + V_1 + r_0 + r_1) - Q/J$. Since $F = 1 + k^2D^2(D-1) \geq D$, we have $V_1 > 2048D > 2048C^2 > Q$; thus $n \geq 0$ and the norm vanishes. This argument retains the computed margin, rather than replacing it by a freely chosen witness.

Exact degree with independent source sizes. Write $r = \deg R$, $n_0 = \deg N$, $c_0 = \deg G_{\mathrm{cap}}$, and let their positive diagonal leading coefficients be ρ, ν, κ. The decisive register degrees are

$$\deg A = r + 4n_0 + 2, \quad \deg C = r, \quad \deg H = 17r + 32n_0 + 19, \quad \deg \Omega = 5r + 20n_0 + c_0 + 12.$$

Exact expansion of $T^2 - qDFG^2$ yields 703 integer monomials in A, C, i. Their exponents are coordinatewise at most $(15, 18, 6)$, with unique simultaneous leading term

$$9 \cdot 2^{18}A^{15}C^{18}i^6.$$

Thus its exact degree is $33r + 60n_0 + 36$. In

$$\Delta_1 = V_1^2 - qa_1 = H^2(T^2 - qDFG^2) + qDF(H^2 - 1),$$

the first term dominates, giving degree $67r + 124n_0 + 74$. In turn $\Omega^2\Delta_1$ uniquely dominates the identity

$$\Theta^2 - \Omega^2(a_0 + qa_1) = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2\{K^2 - 1 + \Delta_1 + 2V_0V_1\}.$$

Its square also dominates the norm's subtracted term. Therefore

$$\deg F_9 = 154r + 328n_0 + 4c_0 + 196, \qquad \mathrm{lc}_{\mathrm{diag}}(F_9) = 81 \cdot 2^{148}\kappa^4\rho^{154}\nu^{328} > 0. \tag{3}$$

This treats the carry block length and the Jones threshold root as distinct quantities; it does not assume $\deg R = 3 \deg N$. The independently expanded defect, modular circuit replays, and exact leading-degree checks audit these algebraic claims. They do not replace the inherited source compiler or Jones ratio theorems, prove consistency, or constitute a new end-to-end proof-assistant certificate.

7.1 Literal graphs and the nine variable tradeoffs

The primary positive polynomial uses positive inputs

$$(b, g, h, s, w, \phi_+, i, j_+, n_+) \in \mathbb{Z}^9_{>0},$$

with $\phi = \phi_+ - 1$, $j = j_+ - 1$, $n = n_+ - 1$. The natural version uses $(b_0, g_0, h_0, s_0, w_0, \phi, i_0, j, n) \in \mathbb{N}^9$ and sets $b = b_0 + 1$, $g = g_0 + 1$, $h = h_0 + 1$, $s = s_0 + 1$, $w = w_0 + 1$, $i = i_0 + 1$. The literal circuit orders these nine slots consistently. Both maps are bijections of the declared domains, and every arithmetic shift is charged.

For each packing and majorant, the JSON records the source certificate, source parameters, domain, ordered inputs, all fixed-leaf recipes, live DAG, named register map, output index, operation counts, and exact-degree certificate. A register name is not an additional unknown. The fixed-binding file connects the recipe leaves to signed or unsigned little-endian payloads; fixed power leaves are materialized too.

The balanced $W = 170$, $p = 169$, $K_c = 4426$ certificate has

$$(r, n_0, c_0) = (18394, 4776, 341),$$

so the quotient-cancelled degree formula gives $4\,400\,764$. Removing its uniform target shift gives $K_c = 4425$ and the odd threshold branch; then $(r, n_0, c_0) = (18390, 4775, 341)$ and the degree is $4\,399\,820$. Its lower degree comes at a higher gate cost because the entire source no longer permits the same homogeneous sharing.

Two other explicitly retained tails use the same norm with different q and positive majorant V_1:

1. The simple tail has $q = 64$ and $V_1 = k(8D^2 - 4D - 1)GH$, proved above.
2. The inherited support tail has $q = 2048^2$ and $V_1 = [T_6 + 256i(k^2 + 2) + 2048]H$. Its uniform majorant proof is retained as a pinned inert dependency.

The quotient-cancelled tail is the one proved in detail in the preceding section. For all three, exact cancellation-sensitive upper degrees are matched by a nonzero top diagonal coefficient. The verifier expands the small defect polynomial, not the huge final polynomial, and propagates its exact degree through the remaining circuit identities. Random modular replays are additional diagnostics, not the sole degree proof.

The finite source verifier proves exact equality of each evaluator with its declared sparse source in the polynomial ring over four independent port indeterminates. In particular, shared homogenization does not assume the ports have special numerical values. Full independent circuit replays then use the actual fixed residues and both domain translations. The primary operation count is never replaced by the cost of an auxiliary certificate graph or a denominator-multiplied polynomial.

8 Ten integers with equal residual weights

The recommended circuit defines a fixed polynomial

$$F_{10}(f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}[f,g,h,k,l,w,x,y,z,m]$$

with ten unrestricted integer unknowns, **554** live binary arithmetic operations, and exact total degree **14,699,152**. Ordinary mathematics formalizable in ZFC, using the identified native compiler and Sun representation theorems, proves

$$\neg\exists \mathbf{u} \in \mathbb{Z}^{10}\ (F_{10}(\mathbf{u}) = 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This neither proves consistency nor claims a universal improvement on Sun's ten-variable theorem. It is not an end-to-end proof-assistant certificate. The prior balanced circuit used 637 operations and degree 34,378,296.

8.1 Fixed source and a calibrated carry bridge

The selective source has 31 natural coordinates and 22 quadratic residuals, with 109 nonzero coefficient records. Only twelve old positive coordinates are shifted by one; positivity of the remaining nineteen follows from the source equations. All four native ports are bound together to the verified 215,492-byte tuple. The already shifted mask port is not shifted a second time.

For the recommended positional certificate, $W = 170$, padding $p = 169$, $D_{\max} = 4086$, and $K_c = 4426$. Write w_i for its 31 coordinate positions and H_j for its 22 targets. With signed source residuals $r_j(\mathbf{n}) = \sum_\alpha p_{j,\alpha}\mathbf{n}^\alpha$, define

$$\begin{aligned} c_{j,\alpha} &= (2-|\alpha|)!\prod_i \alpha_i!\; p_{j,\alpha}, \\ e(\alpha) &= (2-|\alpha|)p + \sum_i \alpha_i w_i, \\ D_c(B) &= \sum_{j,\alpha} c_{j,\alpha} B^{H_j - e(\alpha)}, \qquad\qquad Z_c(B) = \sum_j B^{H_j}. \end{aligned}$$

The independently checked support-isolation certificate proves

$$[B^{H_j}]\big(B^p + \textstyle\sum_i n_i B^{w_i}\big)^2 D_c(B) = 2r_j(\mathbf{n}).$$

It also proves that D_c has positive leading coefficient and $K_c \geq \max H_j + 2$. No full Sidon property is needed.

Let $C_0 = 1 + \sum |c_{j,\alpha}|$. The actual fixed coefficients satisfy $\beta_S = 2^{864026} > 8192C_0$. Define

$$\begin{aligned} b &= 1+3f, & B &= \beta_S b^2, & \mathcal{C} &= bB^W, \\ \ell &= \sum_i B^{w_i}, & N_0 &= B^{W+1}, & N_1 &= B^{K_c+1}, & N &= N_0N_1, \\ M_0 &= N_0 - 1 - (b-1)\ell, \\ T_c &= M_0 + N_0(B-2)Z_c(B), \\ J_c &= 2(B^p+g)^2 D_c(B) + Z_c(B), & S_c &= g + N_0 J_c, \\ X &= N^2 + (S_c+T_c)N + (N-1-S_c), \\ Y &= 2^{13824395}B^{4775}b. \end{aligned}$$

All exponents are fixed nonnegative integers. The final expression for Y is the literal polynomial form of the exact mask-popcount threshold.

Suppose $f \neq 0$, b is a square, and $0 \leq g < 2\mathcal{C}$. Then $b \geq 4$, $B > 2b$, $D_c(B) > 0$, and

$$2(B^p + g)^2 D_c(B) < 18 C_0 b^2 B^{K_c} < N_1/256.$$

The distinct targets and their two-position headroom give $Z_c(B) < 2B^{K_c-2}$ and $(B-2)Z_c(B) < B^{K_c-1}$. Consequently $S_c, T_c \geq 0$ and $S_c + T_c < N$. Thus $X \geq N^2 > 3b$ and $Y > \max(b, 256)$, before any power-of-two recovery. Also $b, B, N_0, N \equiv 1 \pmod 3$, so $X \equiv 1 + T_c \equiv 1 - 22 \equiv 0 \pmod 3$.

If b is also a power of two, the low mask decodes $g = \sum_i n_i B^{w_i}$ with $0 \leq n_i < b$. Write h_t for the coefficients of $(B^p + \sum_i n_i B^{w_i})^2 D_c(B)$. Then $|h_t| < B/8$. The nominal digit of J_c at a target is $2h_t + 1$. Every incoming borrow is -1 or 0; after normalization this digit is 0 or 1 exactly when $h_t = 0$. These are precisely the digits disjoint from the mask $B - 2$. Hence $S_c \,\&\, T_c = 0$ detects all 22 residuals.

Writing $N = 2^u$, the separated binary blocks give

$$\operatorname{popcount}(X) = 1 + u + \operatorname{popcount}(T_c) - c(S_c, T_c),$$

where c is the nonnegative propagated carry count. For $b = 2^k$ and $B = 2^v$, the mask has $\operatorname{popcount}(T_c) = (W + 23)v - 31k - 22$. Legendre's formula $v_2\binom{2X}{X} = \operatorname{popcount}(X)$ now proves

$$Y \mid \binom{2X}{X} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

Conversely, a source zero supplies arbitrarily large $b = 4^q$ and $b \leq g < \mathcal{C}$: the unshifted coordinates D, Δ are both positive, and at most one has position zero. This supplies exactly the inherited Sun source interface. The sparse-padding and popcount mechanisms are adapted from the eighth incoming archive; the complete interior-padding proof is in `SPARSE_SOURCE_PROOF.md`.

8.2 Complete polynomial tail

Using the preceding $b, X, Y, \mathcal{C}$, form these polynomial abbreviations:

$$\begin{aligned}
L &= lY, & U &= 2LX, & V &= 4gwY,\\
A &= U(V+1), & B_1 &= 2X+1, & C &= B_1 + (A-2)h,\\
P_n &= U^2 V, & K &= X + 1 + k(P_n - 2), & S &= 2A - 5,\\
T &= 3bwC - 2((bw)^2 - 1),\\
\Delta &= 8\mathcal{C}^3 g K^2 - g^2\{32(C - KL)^2\mathcal{C}^3 + g^2 K^2\},\\
O &= f^2 l^2 \Delta, & \xi &= zx,\\
D &= (A^2-4)C^2 + 4, & E &= C^2 D\xi,\\
F &= 4(A^2-4)E^2 + 1, & G &= 2(A+2)(A-2)^2E^2 - 1,\\
H &= C + (B_1 - C)F + 16EFy, & I &= (G^2-1)H^2 + 1,\\
r_0 &= b\{1 + (3O-4)b^2 z^2\}, & r_1 &= (P_n^2 - 4)K^2 + 4, & r_2 &= DFI,\\
\Sigma &= \xi S^5, & t &= T + Sy,\\
\Theta &= (t^3 - tS^2 - S^3)(t^2 + 23S^2).
\end{aligned}$$

The new weights are $(1, b, b)$. Set

$$Z = \Sigma m - \Theta, \qquad a_0 = \Sigma^2 r_0, \quad a_1 = \Sigma^2 b^2 r_1, \quad a_2 = \Sigma^2 b^2 r_2,$$

and define the final polynomial by

$$Q = Z^2 + a_2 - a_0 - a_1, \qquad R = Q^2 + 4Z^2a_2 - 4a_0a_1, \qquad F_{10} = R^2 - 16Z^2a_2Q^2.$$

Every auxiliary letter is an abbreviation, not an unknown. The exact circuit file supplies the positional data, fixed leaves and binary gates.

8.3 Why the smaller weights are valid

The norm identity is

$$R^2 - 16Z^2a_2Q^2 = \prod_{\epsilon_0,\epsilon_1,\epsilon_2\in\{\pm1\}} (Z + \epsilon_0\sqrt{a_0} + \epsilon_1\sqrt{a_1} + \epsilon_2\sqrt{a_2}).$$

It holds over the complex numbers, including dependent squareclasses and zero radicands. Since S is odd, it is nonzero. The quintic $(v^3 - v - 1)(v^2 + 23)$ has no rational root, so $\Theta \neq 0$. At $\Sigma = 0$ the norm is Θ^8, excluding that branch. Monicity of the normalized norm and the rational-root theorem imply $\Sigma \mid \Theta$, and the product identity gives a rational signed sum

$$\pm\sqrt{r_0} \ \pm b\sqrt{r_1} \ \pm b\sqrt{r_2}.$$

No positivity has yet been assumed.

Put $J = 1 + (3O - 4)b^2z^2$; then $\gcd(b, J) = 1$. If $|b| > 1$ and $r_0 = bJ$ were in a nontrivial signed squarefree class d, write $r_0 = da^2$. Linear independence of distinct squarefree radical classes would force $b \mid a$: every other contribution in this class is a multiple of b. This would imply $b \mid J$, a contradiction. The exceptional case $|b| = 1$ is $b = 1$, hence $f = 0$ and $r_0 = 1 - 4z^2$ is odd. Both other radicands are divisible by four, so an odd squareclass coefficient cannot be canceled by their even coefficients. Thus r_0 is a nonnegative square in every case. It cannot vanish: $J = 0$ forces $b = 1$ and $O = 1$, inconsistent with $f = 0$. The actual $b = 1$ case is then impossible because $z \neq 0$.

Unique factorization and $b \equiv 1 \pmod 3$ make b a positive square. Since $J > 0$ and $z \neq 0$, one has $O > 0$, hence $f, l \neq 0$ and $\Delta > 0$. Now $\mathcal{C} > 0$. The formula for Δ first excludes $K = 0$ and $g \leq 0$, and then yields $g < 2\mathcal{C}$. The source size bounds apply, so $|L| \geq 2$, $|U| \geq 4$ and, as $V \in 4\mathbb{Z}$, $|A| \geq 4$. Furthermore

$$K^2 > 4g(C - KL)^2 \quad \Longrightarrow \quad |C| > |K|(|L| - \tfrac{1}{2}) > |K|.$$

For $V \in 4\mathbb{Z}$, $|V| \leq (V + 1)^2$, whence $|P_n| \leq A^2$ and

$$r_1 < A^4C^2 + 4.$$

In contrast, C is odd, $D \geq A^2 \geq 16$, and $E^2 = C^4D^2\xi^2 \geq C^2$. Thus $F \geq A^2C^2 + 1$. Since G is odd, $I \geq 1$, so

$$r_2 = DFI \geq A^4C^2 + A^2 > A^4C^2 + 4 > r_1.$$

This strict separation is established before assuming either remaining square predicate. Subtracting the integral first root and dividing by b leaves a rational sum $q = \pm\sqrt{r_1} \pm \sqrt{r_2}$. It cannot be zero because $r_1 \neq r_2$. Squaring gives $\pm\sqrt{r_1} = (q^2 + r_1 - r_2)/(2q)$; both roots are therefore rational, hence integral. Negative candidate radicands and dependent classes have been resolved, not assumed away.

We recover the same three square predicates and divisibility as the inherited finalizer. Conversely those predicates provide an integral signed weighted root sum and hence a suitable integer m. This proves the exact existential equivalence of the replacement.

8.4 Circuit and exact-degree certificates

The circuit shares a homogeneous evaluation of the positive parent, with only the declared twelve coordinate shifts and diagonal corrections for factorial weighting. Negative row signs reverse a subtraction. A hybrid alternative uses sparse Horner evaluation only for rows whose homogeneous offsets would be negative. Exact expansion matches all 115 resulting monomials over four independent native-port indeterminates.

Further exact gate savings use, with $a = A/2$ constructed without division,

$$G + 1 = (a - 1)(F - 1), \qquad G^2 - 1 = (G + 1)(G - 1),$$
$$T = 2 + bw(3C - 2bw), \qquad H = C + F\{16Ey - (A - 2)h\},$$
$$R = Q^2 + 4(Z^2 a_2 - a_0 a_1).$$

The live gate count is 347 multiplications, 149 additions and 58 subtractions. The propagated upper degree is 14,699,152. Substituting all ten variables by T gives a nonzero coefficient 982070748 modulo 1000000007 at that degree, proving exactness. Two additional primes independently give nonzero top coefficients.

A 513-operation alternative has exact degree 25,838,224; a 558-operation hybrid has degree 14,696,016. These are search results, not minimality claims. The independent verifier also replays every source and tail, checks fixed-byte hashes, and checks exact ring identities. Finite regressions include 38,016 strict-separation cases and negative controls showing that $(1, b, b)$ is not an unconditional square detector. The mathematical proofs, not the finite samples, establish universality.

9 A constant-cancelled positive norm minimum

The short-equation frontier gains a fifteen-positive-variable polynomial of degree 64 evaluated in 72 binary arithmetic operations, and a 70-operation degree-80 version. The inherited 66-operation, degree-157 formula remains the operation-count comparison. All four ports are rebound together to the verified native tuple with base 2^{429959}; the high mask is already shifted.

For the 72/64 version the positive inputs are $J, F, \alpha, z, f, h, i, T, s, w, r, \eta, \zeta, u, \gamma$. Define

$$q = \beta J + 1, \qquad X = wq, \qquad Y = sq^3, \qquad E = XY,$$
$$k = \eta + \zeta, \qquad c = kY + \eta, \qquad a = E + Y, \qquad H = 4a + 3.$$

$$\Delta = a^2 + H, \qquad D = X + ac + \gamma H, \qquad \ell = EkY,$$
$$V = f^2(cT - R) - c, \qquad I = ic^2, \qquad A_0 = \Delta(f^2 - 1).$$

$$R = ((q - 1)(q - F) + \alpha)(q^2 - 1) + (M_C + qM_F)J.$$

$$N_0 = r^2 + \ell(2r - k), \qquad N_1 = D^2 - \Delta c^2,$$
$$N_a = (V + u)^2 - A_0 u(2V + u), \qquad N_k = k - hE - R.$$

$$N_t = (K_0 + w)(q - F - \alpha) + (q - F) - z(q - 1).$$

Put $A = N_0 N_1$, $B = N_a N_k N_t$, and $C = I^2 - A_0$. The equation is

$$A^2 + B(B - 2) + C^2 = 0. \tag{S64}$$

All displayed auxiliary letters denote polynomial abbreviations, not new witnesses. The literal division-free DAG uses 40 multiplications, 18 additions and 14 subtractions, including all literal operations.

Soundness. The first and main norms cannot vanish on the positive domain: their discriminants $XY^2(XY^2+1)$ and $(a+2)^2-1$ are nonsquares. Hence $A^2 \geq 1$. Completing the square in (S64) gives $A^2+(B-1)^2+C^2=1$, so $A^2=1$, $B=1$, $C=0$. The consecutive-product negative-Pell obstruction and the modulo-four main-norm obstruction give $N_0=N_1=1$. Now $A_0=I^2>1$ and the auxiliary norm has square residue modulo four, giving $N_a=1$. Restore $\tau=\ell+r>0$ and $y=|V+u|>0$; the latter cannot vanish since its norm is one. Set $T_{\text{old}}=fT$. The resulting tuple solves the inherited ordinary product equation, with $N_kN_t=1$. Its signed strong-rank theorem recovers both unit signs, including the weak transport boundary, and extracts the blank K215 computation. No division of a supplied multiplier by Δ is made.

Positive completeness. For a canonical normalized compiler witness, $c=\psi_{a+2}(R)$ and $f^2=1+\Delta j^2c^4$. Choose $i=\Delta j$. The verified native low-mask phase gives $R\equiv 3 \pmod 4$. For the odd-index Pell polynomials,

$$\chi_t(2v+1)=tQ_v(t^2), \quad Q_v(0)=(-1)^v(2v+1), \quad Q_v(1-(a+2)^2)=(-1)^v\psi_{a+2}(2v+1).$$

With $S=ic^2$, choose $V=\chi_S(R)/S$ and $y=\psi_S(R)$. Since $S^2=\Delta(f^2-1)$, these identities give $V\equiv -c \pmod{f^2}$ and $V\equiv -R \pmod c$. Also $f^2\equiv 1 \pmod c$, so

$$T=\frac{V+c+Rf^2}{cf^2}$$

is a positive integer. Moreover $V\geq 4S^2-3>1$ and the auxiliary norm implies $y>V$, giving positive $u=y-V$; the first norm similarly gives positive $r=\tau-\ell$. Thus $A=B=1$, $C=0$. This reconstructs auxiliary coordinates; it does not assert divisibility of every old quotient by f.

Exact degree. Let $q_0=\beta J$, $k_0=\eta+\zeta$, $E_0=wsq_0^4$, $c_0=k_0sq_0^3$, and $T_0=w(q_0-F-\alpha)-zq_0$. The degrees of A,B,C are $30,32,22$, respectively. The unique degree-64 term is B^2, with leader

$$4h^2E_0^6c_0^2T^2f^8u^2T_0^2.$$

The coefficient of $J^{32}h^2\eta^2w^6s^8T^2f^8u^2z^2$ is $4\beta^{32}\neq 0$. Without either gap the same finalizer has 70 gates and degree 80. With both gaps but old $V=c(Tf-1)-Rf^2$, it has 73 gates and degree 62. The target cancellation $(B-1)^2-1=B(B-2)$ saves one operation.

The independent checker verifies full DAG liveness, all four native payload hashes, 1200 direct-formula evaluations, 72 complete finite-field univariate degree replays, and five exact Pell reconstruction examples. Normal and optimized runs agree. These checks corroborate the unbounded proof; inherited compiler and rank theorems remain dependencies, not new kernel-certified results. No consistency truth value or global optimality is claimed.

9.1 All short family domains and literal variants

Every short family polynomial has fifteen strictly positive inputs. The 70/80, 68/101 and inherited 66/157 graphs use τ, y instead of r, u; the 71/78 graph uses r, y; the 72/64 and 73/62 graphs use r, u. Other input names are exactly those listed above. The 73/62 version retains the old quotient register $V=c(Tf-1)-Rf^2$, whereas the squared-modulus versions use $V=f^2(cT-R)-c$. They are existence-equivalent through constructive witness rebuilding, not equal polynomials on arbitrary input tuples.

The complete root-directory short frontier is

Operations	Multiplications	Additions	Subtractions	Exact degree
66	38	15	13	157
68	38	16	14	101
70	39	17	14	80
71	40	17	14	78
72	40	18	14	64
73	41	18	14	62

The incoming 75-operation degree-44 form remains another nondominated fifteen-positive tradeoff; the table lists this delivery's family, not the entire catalogue.

The 66-operation formula is inherited and rebound to the whole native tuple, not a new formula. The other entries are explicit new compositions or simplifications. The supplied degree proof records every top homogeneous factor, and complete finite-field univariate replay verifies the claimed exact degrees for the actual coefficient binding.

For the 68-operation product version define $N_s = 1 + I^2 - A_0$ and use $N_0N_1N_aN_kN_tN_s - 1$. The factor N_s is a unit at a zero and cannot be -1 modulo four, so $N_s = 1$ and the preceding proof applies. Its degree is $22 + 18 + 30 + 7 + 2 + 22 = 101$, attained because all factor leaders are nonzero. For the inherited 66-operation version set $Q_0 = (\Delta ic^2)^2$ and use

$$N_0N_1\{Q_0(V^2 - y^2) + y^2\}N_kN_t(\Delta f^2 - Q_0) - \Delta.$$

Its direct ordinary bridge takes $i_{\text{ordinary}} = \Delta i$ and $T_{\text{old}} = fT$ after unit extraction. Thus no missing historical coefficient payload is needed to interpret the active instance.

The squared quotient does *not* directly remove a radical from the previous eleven-/twelve-positive equations: it removes the occurrence that coupled the strong square to the auxiliary Pell norm. The following explicit Pell-fiber test isolates that failure. Take $a_0 = 3$, $\Delta = 8$, $R = 3$, $c = 35$, $i = 1$, $S = 9800$, and

$$D_{\text{strong}} = 12005001, \quad T = 1, \quad V = 384159997, \quad y = 384159999.$$

The main Pell root is 99. Then D_{strong} is nonsquare, but $V = D_{\text{strong}}(cT-R)-c$ and $S^2V^2-(S^2-1)y^2 = 1$. This is not asserted to be a compiler zero; it shows exactly why that naive radical deletion is invalid. The new short frontier therefore makes no new eleven-/twelve-variable claim.

10 Reproduction and verification boundary

10.1 One command from a clean extraction

Run `python3 verify_release.py` from the extracted release, or invoke that script by absolute path from any directory. Python 3.10 or later, SymPy and NumPy are required. The top-level verifier rejects optimized Python (`-O` or `-OO`), validates the exact sealed file set and all SHA-256 hashes, and checks the retained direct-source Git blob identities. It also checks the selected archive-member hashes against the recorded provenance. It then makes an isolated temporary copy and runs only this release's authored checks there. Any failed check exits nonzero; the sealed original is checked again afterward.

The eight original incoming archives were authenticated at intake but are not included. Clean replay therefore does *not* recompute their Git blob hashes from nonexistent archive bytes. The retained ledger records those intake results and binds the selected member bytes. Genuine direct Git-blob checks apply to the included repository-source files, separately identified in the provenance ledger. These are integrity

checks, not an external digital signature: the manifest cannot authenticate its own origin. The recipient should record the outer ZIP digest through a trusted channel.

The optional `rebuild_artifacts.py` regenerates 112 source, packing, circuit and fixed-payload files in a temporary copy and compares their exact bytes; this succeeded in the recorded Python/SymPy environment.

The release contains editable LaTeX and the rendered PDF, explicit arithmetic DAGs, exact fixed coefficient payloads and recipes, source and packing JSON, mathematical proofs, inherited licenses, newly authored builders and independent verifiers. No input ZIP duplication, search binaries, font cache, or transient computation logs are needed. The `negative_controls.py` script tests rejection of optimized execution, changed/missing/unexpected files, corrupted source bytes even after rehashing the manifest, and a malformed positional certificate.

For a standard TeX Live or MiKTeX install, run `pdflatex report.tex` twice from `report/`, or `latexmk -pdf report.tex`. The Linux helper uses the existing research environment's TeX files, without shell escape or a new installation. Compile in a working copy, since a sealed manifest intentionally rejects newly generated auxiliary files.

10.2 What the finite checks establish

1. Reconstruct the coherent native tuple from the declarative mode formulas and inert transition table; check all finite decoder cases, controller residues, payload sizes and hashes.
2. Reconstruct all 22 parent residuals, apply exactly the twelve shifts, verify all 109 factorial-weighted records and 62 active monomials, the nonzero-origin guard and actual coefficient bounds.
3. Exhaustively test the 109-by-528 source/monomial isolation condition for each of nine packing certificates, including row signs and interior padding; verify positive leading coefficient and target headroom.
4. Test the exact complement/popcount identity, signed borrow cases, threshold parity branches and modulo-three conditions. These finite regressions accompany, rather than replace, the universal proofs.
5. Prove exact symbolic equality between every shared or sparse nine-variable source evaluator and its declared source polynomial. Independently reconstruct the small radical-defect polynomial and degree dominance; replay complete literal graphs and both domain translations with actual coefficient residues.
6. For the integer family, check all 24 literal graphs, source identities, fixed payloads, zero-scale and modulo-three cases, norm identities and three nonzero top-degree residues. Matching degree upper bounds and nonzero diagonal coefficients certify the exact total degrees.
7. For the short positive family, traverse every DAG for liveness and paid operations, independently evaluate the formulas, replay full univariate polynomials over three finite fields, and test the exact Pell reconstruction. A separate review audits the universal sign, positivity and phase arguments.

A numerical test passing on finitely many assignments is not a universal compiler theorem. Exact symbolic ring identities and nonzero modular coefficients do have the finite mathematical force explicitly stated. The ordinary universal proofs, composed with the inherited semantic contracts, establish the consistency equivalences formalizable in ZFC.

10.3 Later formalization status at a separate pin

The source pin for this construction remains `4b4349064a750ba6347293b55874a7f92abdd0db`. A separate, later snapshot at `d2f19e4d5cc4030f60049333285eea86a1ef401c` records that wave 7 was placed as collection entries 46–53. More importantly, its status documents report a closed executable actual-ZFC

consistency equation: the concrete arithmetic refutation-program extraction and supplied-data bridge are no longer open. The project also reports checked generic simultaneous-quadratic coefficient extraction and a multi-target central-binomial carry test under explicit packing bounds.

The same documents distinguish remaining work: the report-specific ordinary P67 compiler is not a newly checked Lean proof; a smaller closed quadratic-to-Jones adapter is not supplied by those generic lemmas; and the internal object-level ZFC consistency equivalence remains open, despite checked internal arithmetic and polynomial-side sentence construction. The ordinary external equivalence and an internal derivation are different claims. We have not rerun the later Lean build here, and its progress does not retroactively kernel-certify these new literal equations.

11 Further research directions

1. Optimize actual gate count and exact final degree jointly with the source support, row signs, padding position and target shifts. A smaller maximum weight can lose to a layout permitting shared homogeneous evaluation. The nine small isolation certificates supply a stable proof-output format.
2. Seek a still lower polynomial defect in $T^2 - qDFG^2$, rather than a numerically tighter square-root estimate alone. Any improvement must prove strict positivity before squarehood or divisibility is known and must certify the primary equation after leading cancellations.
3. Extend the twelve-coordinate positivity recovery by proving a further unguarded coordinate positive from the exact source equations. An apparent support saving without a two-way domain proof is not an improvement.
4. The squared-modulus quotient removes a square-root occurrence needed by the previous eleven-/twelve-variable radical norm. Find a replacement coupling rather than silently deleting that strong-square condition. The included explicit Pell-fiber counterexample rules out the naive inference.
5. Connect the closed, smaller simultaneous-quadratic adapter to the current executable compiler and then to an object-level ZFC derivation. The later pinned project progress is relevant infrastructure, not a kernel certificate of the literal polynomials delivered here.
6. Prove finite-catalogue or restricted-family lower bounds for circuit cost, target placement, or unknown count. No present heuristic search gives a global simplicity theorem.

These are specific open directions, not conditions needed to specify or replay the present equations.

References and provenance

1. ProveIt incoming snapshot: eight exact wave-7 arrivals at commit 4b4349064a75. The archive and selected-member ledgers identify every borrowed proof.
2. ProveIt construction baseline: the pinned construction collection at commit 124c07c0b77a. The retained source blobs include the quadratic-parent proof, native compiler proof, ordinary signed compiler and arithmetic interfaces.
3. J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* 47 (1982), 549–571. `https://doi.org/10.2307/2273588`. The precise inherited interface is retained in `inherited/nine/NineRatio.lean`. No fresh compilation is claimed.
4. Zhi-Wei Sun, Further results on Hilbert's tenth problem, `https://arxiv.org/abs/1704.03504`, version 7 (2021), and the later integer bridge cited by the pinned source, `https://arxiv.org/abs/2609.01594`, version 2 (2026). The mathematical contract used here is `inherited/sun/SIGNED10_PROOF.md`, Sections 4–8.
5. Later formalization-status snapshot: commit d2f19e4d5cc4. Exact documents and Git blob pins are in `review/postpin/`, separate from the construction sources.